A Hybrid Quantum-Classical Language Model for Text Generation: A Case Study on The Gift of the Magi

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Abstract—This paper presents a hybrid quantum-classical language model for text generation, evaluated through a detailed case study of O. Henry's classic short story, The Gift of the Magi. The model integrates a classical transformer with a quantum generator to enhance the extraction of abstract narrative features. Our system architecture includes stages for input tokenization, contextual embedding through multi-head attention, quantum- enhanced feature extraction using a parameterized quantum circuit, and output refinement through a classical discriminator. Quantitative evaluations using the BLEU score, besides quali- tative analysis, reveal that while the hybrid approach generates syntactically coherent text, it currently falls short in capturing the nuanced symbolic and emotional depth of the original narrative. The findings show the present limitations of quantum-enhanced language models and point towards future research directions in refining quantum circuit designs, reducing noise, and improving integration strategies to fully exploit the potential of quantum computing in natural language processing.

I. Introduction

Text generation is a key NLP (natural language processing) task with many use cases, including machine translation, dialogue generation, creative writing, text summarization, etc. It is vital for enhancing human-computer interaction and streamlining content production to have a model capable of generating coherent, context-appropriate, and syntactically correct text. In recent years, deep learning-based methods have made rapid progression in the world of text generation, through generating language models like GPT [3] (Genera- tive Pre-trained Transformer), BERT (Bidirectional Encoder Representations from Transformers), T5 (Text-to-Text Transfer Transformer) [2]. They work with large datasets and complex self-attention mechanisms to produce incredibly realistic text, often brushing the line with human-created content [4].

Despite their outstanding performance, classical transform- ers have limitations when it comes to generating deeply symbolic, emotional, and nuanced written text. Most models depend more on statistical patterns and syntactic structures rather than a genuine understanding of abstract themes. So, as good as they are at generating factual and task-specific text, replicating the nuanced narrative arcs, symbolism systems, and overarching themes of literature has proven to be a chal- lenge. This limitation arises from the fact that they are built based on deterministic training paradigms, which reflect an extrapolation of meaning from large quantities of data without necessarily embodying a deep conceptual understanding.

To overcome those challenges, the capability of quantum computing to improve the tasks of natural language processing was investigated. Quantum computing based on principles like ”superposition”, ”entanglement”, and ”decoherence” have shown promise where it can provide alternative data represen- tations in classical models. Quantum circuits can represent and manipulate information in fundamentally different ways than classical systems, possibly allowing for a more sophisticated representation of language [11]. In this paper, we investigate the integration of quantum computing into text generation by introducing a hybrid quantum-classical model for processing and generating text.

A. Quantum-Assisted Language Models

Quantum-assisted language models, with their ability to flexibly represent intricate patterns, offer a new paradigm for language generation along with innovations such as early exit for termination. Quantum superposition allows them to perform many computations at once, in contrast to classical models, which use classical logic gates to process information sequentially or in parallel [5]. Here we are introducing the possibility of nonlinear representations, which we believe can contribute to modeling high dimensions of abstract storytelling properties like irony, allegory, and sensibility.

However, we are still in the early stages of quantum computing. Current hardware is limited by qubit coherence time, noise, and other constraints on scalability. Due to these constraints, most quantum NLP research today adopts a hybrid approach, where quantum models act as helper nodes in classical architectures. In this study, we investigate combining such benefits through a hybrid approach using a quantum generator in conjunction with a classical transformer-based model to generate informative representations of literary text.

B. Case Study: The Gift of the Magi

In our experiment we use O. Henry's classic short story, The Gift of the Magi, a work of literature rich in emotional content, situational irony, and symbolism [18]. The story revolves around a young couple, Jim and Della, who each sacrifice their most treasured possessions to purchase each-other a gift, only to find out that their sacrifices make the gifts impractical. This story covers elements of love, sacrifice, and selflessness, making it a great story to evaluate how well quantum-assisted language models can get deep literary elements. Key reasons for selecting The Gift of the Magi include:

- Conciseness & Narrative Structure – The short length of the story allows for focused evaluation without the complexities of processing an extensive dataset.
- Symbolism & Thematic Complexity – The story's ironic twist, emotional undertones, and layered

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meanings make it a challenging test case for any text generation model.

- *Well-Established Human Summaries* – The story has been extensively studied, providing a robust reference set for evaluating model-generated text.

II. Model Architecture And Methodology

The proposed hybrid Quantum Generative Adversarial Net- work (QGAN) model for text generation integrates both clas- sical and quantum computing paradigms. The model consists of three primary components:

1) A classical transformer, responsible for generating con- textualized word representations.
2) A quantum generator, that leverages a parameterized quantum circuit to capture abstract linguistic features [6].
3) A classical discriminator, that evaluates the authenticity of the generated text and helps refine the model's outputs.

Together, these components create a hybrid architecture that aims to enhance the generation of complex literary text, such as The Gift of the Magi.

A. Classical Transformer Backbone

The transformer architecture is a key component of modern natural language processing (NLP) models, providing efficient parallel processing and capturing long-range dependencies between words. In our model, the classical transformer back- bone is responsible for learning contextual relationships and generating probability distributions over vocabulary tokens. Key Components of the Transformer Backbone include:

Token Embedding: In The Gift of the Magi, each character is mapped to a high-dimensional vector in an embedding space. It enables the model to learn distributed representations for tokens, meaning that words with similar meanings or contexts end up having similar embeddings in the vector space. More formally, for vocabulary size V and embedding dimension d, the embedding matrix E is defined as:

$$E \in R^{V \times d}. \qquad (1)$$

Each word wi is mapped to a vector E(wi), enabling the model to learn distributed representations.

Positional Embedding: Unlike recurrent neural networks (RNNs), transformers do not process sequences in order. To introduce sequence information, positional encodings are added to the token embeddings. The positional encoding for a word at position t is computed as:

$$PE(t, 2i) = \sin\left(\frac{t}{10000^{\frac{i}{d}}}\right), \qquad (2)$$

$$PE(t, 2i+1) = \cos\left(\frac{t}{10000^{\frac{i}{d}}}\right), \qquad (3)$$

where, i indexes the embedding dimensions.

Multi-Head Self-Attention: The transformer uses multi-head attention to compute dependencies between words, allowing the model to attend to different parts of the input simulta- neously [1]. The scaled dot-product attention mechanism is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \qquad (4)$$

where, Q, K, and V represent the query, key, and value matrices, respectively, and dk is the key dimension.

Feedforward Network (FFN): Each attention layer is fol- lowed by a fully connected feedforward network:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2, \qquad (5)$$

where, W1, W2 are weight matrices and b1, b2 are bias vectors. Layer Normalization & Residual Connections: Layer nor- malization and residual connections ensure stable training andprevent vanishing gradients:

$$\text{LayerNorm}\,(x + \text{FFN}(\text{Self-Attention}(x))) \qquad (6)$$

Mathematical Representation of Transformer Output: For an input sequence X, the transformer outputs a hidden represen- tation H, which is then mapped to logits L using a language modeling head:

$$L = W_{lm} \cdot \text{LayerNorm}(H), \qquad (7)$$

where, Wlm is the learnable weight matrix.

B. Quantum Generator

The quantum generator is a core innovation of our model, designed to enhance feature representations by leveraging quantum states and entanglement [7]. This component pro- cesses input embeddings using a parameterized quantum cir- cuit (PQC) and outputs abstract features that encode linguistic complexity beyond classical approaches [8]. Key Components of the Quantum Generator include:

Quantum Circuit:

- A Hadamard gate (H) is applied to all qubits to create a superposition state, allowing the quantum system to explore multiple possibilities simultaneously.
- The TwoLocal ansatz is applied, consisting of parame- terized rotation gates (RY, RZ) and entanglement layers (CZ gates) to introduce correlations between qubits [10].

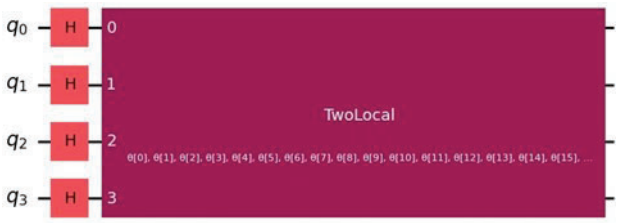


Fig. 1. Proposed Circuit Design using TwoLocal Ansatz

Quantum State Preparation: The input word embedding x is encoded into a quantum state $|\psi\rangle$:

$$|\psi\rangle = U(x)|0>, \qquad (8)$$

where, U (x) is a parameterized unitary operation [12].

Feature Extraction: After quantum computation, the quan- tum measurement produces an output vector z, formulated as:

where, θ are the trainable quantum parameters.

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$$z = f_Q(x; \vartheta), \quad (9)$$

Quantum-Classical Integration: The quantum generator is integrated into PyTorch via TorchConnector, enabling end-to-end training with classical optimizers [14].

C. Classical Discriminator

A classical discriminator is used to assess the quality of text generated by the quantum-enhanced model. The discriminator is a simple feedforward neural network (FNN) that predicts whether a given text is real (authentic text from The Gift of the Magi) or synthetic (generated by the quantum generator) [15]. Structure of the Discriminator comprises of:

- Input Layer: Takes feature representations from the quantum-classical model.
- Hidden Layers: Applies non-linear transformations using ReLU activation.
- Output Layer: Uses a sigmoid function to output a probability score between 0 (fake) and 1 (real).

Mathematically, the discriminator function is defined as:

$$D(z) = \sigma(W_3 \cdot \mathrm{ReLU}(W_2 \cdot \mathrm{ReLU}(W_1 z + b_1) + b_2) + b_3), \quad (10)$$

where, W_1, W_2, W_3 are weight matrices, b_1, b_2, b_3 are bias terms, and σ is the sigmoid activation function.

The discriminator is trained using binary cross-entropy loss:

$$L_D = -\mathbb{E}[y \log(D(z)) + (1 - y) \log(1 - D(z))], \quad (11)$$

where, y is the ground truth label.

D. Overall Model & Training Objective

The Quantum Generative Adversarial Network (QGAN) is trained using an adversarial setup, where the generator and discriminator compete:

- The quantum generator tries to produce text indistinguish- able from real data [16].

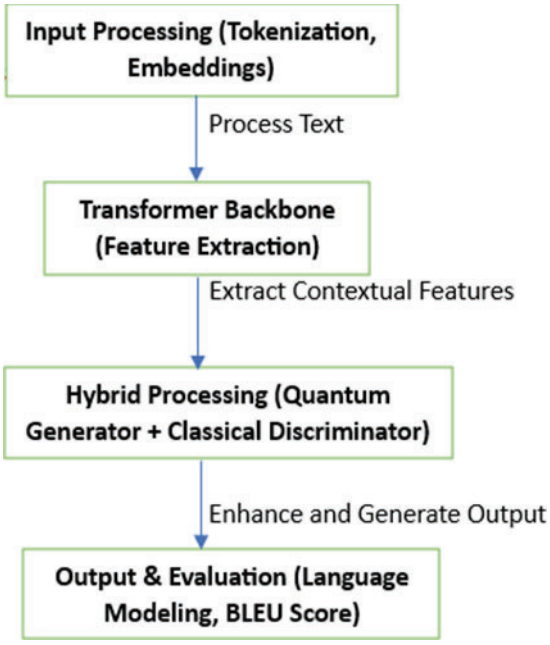


Fig. 2. Workflow of the Procedure

- The classical discriminator attempts to classify real and generated text accurately.

Training Objective: The generator is trained to minimize the cross-entropy loss:

$$L_G = -\sum_i t_i \log L_i, \quad (12)$$

where, ti are the target tokens from The Gift of the Magi and Li are the predicted logits from the quantum-classical model. The training alternates between optimizing LG (for the generator) and LD (for the discriminator), ensuring that the

generated text improves over time.

III. Training Procedure

QGAN Model Training is divided into several structured stages so that both of the model parts are fine-tuned to generate text efficiently (classical transformer backbone plus quantum generator). Since it is a work of considerable metaphor and feeling, one would need to train a sentence generator of this nature on data that allowed it to produce coherent, sensical, relevant words in the same tone as the source. Here, we describe the methodology, the pipeline for preprocessing the data, and the hyperparameters as well as the loss with their evaluation to monitor how well the model is performing and optimize to reach the best model.

A. Training Dataset and Preprocessing

The dataset for training consists of the full text of The Gift of the Magi. Since this is a short-form literary text, careful preprocessing is required to structure the data effectively for training a language model. The following preprocessing steps are employed:

TABLE I. Hyperparameter settings for training.

Hyperparameter	Value	Description
Batch Size	32	Number of text segments processed per training step Block Size
Training Iterations	3000	Total number of training steps Evaluation Iterations Every 50 steps Frequency of validation loss estimation
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- Input Tokenization: As the dataset comprises only one literary work, the text needs to be tokenized at the character level instead of at the word level because of the small number of vocabulary. A dictionary that maps each unique character to its index is created.
- Vocabulary Construction: The vocabulary consists of unique characters from the story. The size of the vocab- ulary indicates how many different embeddings exist in the transformer.
- Sequence Batching: We consume the text in overlapping sequences of 128 tokens (termed the block size). The length of each training batch is 32 sequences (batch size = 32).

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- Training and Validation Split: 80% of the text is used for training, and 20% is reserved for validation. This split allows for tracking generalization performance over time.

After preprocessing, the dataset is converted into numerical tensors for efficient processing using the PyTorch framework.

B. Hyperparameter Selection

Hyperparameters play a crucial role in determining the stability and convergence of training. The following values were empirically chosen based on prior experiments with language models:

C. Training Loop and Optimization Strategy

The training loop is designed to iteratively update the model parameters while minimizing loss and ensuring convergence. The process follows these key steps:

1) Sampling Mini-Batches of Text: Random sample 32 se- quences from the training set. All sequences are padded (if needed) to give all inputs the same length. The input batch converts its format to tensor and moves it to available hardware (GPU or CPU).

2) Forward Pass through the Model: The tokenized se- quences are forwarded through a stacked Transformer Backbone to obtain contextualized hidden representa- tions. The output is split:

- One branch is processed classically using standard transformer layers.
- The other branch is passed into the Quantum Gen- erator, which maps the embeddings onto a quan- tum state, processes using parameterized quantum circuits, and extracts a quantum-enhanced feature representation.

3) Computing the Loss: The loss function used for training is the Cross-Entropy Loss, defined as:

$$L = -\sum_{i} t_i \log L_i, \qquad (13)$$

where, ti are the ground truth target tokens and Li are the predicted logits from the model.

4) Backpropagation and Model Update: Gradients are com- puted using automatic differentiation. The AdamW op- timizer updates model weights using the update rule:

$$\vartheta \leftarrow \vartheta - \eta \nabla_{\vartheta} L, \qquad (14)$$

where, θ represents the model parameters, η is the learning rate, and θL is the computed gradient.

5) Estimating Validation Loss: Every 50 iterations, a batch is sampled from the validation set. The model generates text without updating weights, and the validation loss is computed to assess model generalization.

D. Convergence Monitoring and Training Stability

To ensure effective learning and to prevent overfitting, the following strategies are implemented:

- Loss Visualization: A loss curve is plotted over time to track learning progress. Early stopping is applied if validation loss stagnates or increases.
- Gradient Clipping: To prevent unstable updates, gradient clipping is enforced as:

$$g = \min(g, \text{threshold}), \qquad (15)$$

ensuring that updates remain within a controlled range.

- Evaluation of Generated Text: The model periodically generates sample sentences, allowing for qualitative as- sessment of text coherence and fluency. A manual review helps verify that the generated text captures O. Henry's style and themes.

E. Challenges in Training Quantum-Assisted Models

Training hybrid quantum-classical models introduces unique challenges [19]:

- Circuits Noise: Current quantum hardware adds mea- surement noise and decoherence to the model structure, impacting the retrieving process of quantum-extracted features. Simulated quantum environments (e.g. Qiskit's AerSimulator) are used for this purpose to mitigate this.
- Computational Bottlenecks: Quantum simulations intro- duce overhead costs, and models with quantum com- ponents exhibit larger training times when compared to purely classical models. The transformer backbone is pretrained before the integration of the quantum generator to speed up training.
- Quantum Gradient Stability: Quantum updates to param- eters can be noisy by nature, resulting in training insta- bility [9]. To address this challenge, hybrid optimizers and dynamic tuning of the learning rates assigned to the quantum components are utilized.

IV. Evaluation Via Bleu Score

It is important to evaluate the performance of a text genera- tion model in terms of how well it captures linguistic structure, meaning, and fluency. Using the Bilingual Evaluation Under- study (BLEU) Score, we quantitatively evaluate the quality of text generated by our Quantum Generative Adversarial Network (QGAN) with respect to the original text used, which is The Gift of the Magi.

BLEU is a popular metric for evaluating machine translation and text generation, measuring the n-gram precision of generated text compared to a reference corpus. BLEU score is a metric that represents the similarity to the reference texts in terms of syntax and semantics, with high values relating to high similarity and low values showing more divergence from the desired style or meaning.

A. Mathematical Definition of BLEU Score

The BLEU score is defined as:

$$\text{BLEU} = BP \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \qquad (16)$$

where,

- c is the length of the candidate (generated) text.

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- r is the length of the reference text.

If the generated text is shorter than the reference text ($c < r$), the penalty reduces the BLEU score, discouraging incomplete or truncated outputs.

B. Evaluation Setup for The Gift of the Magi

To assess our QGAN model we calculate BLEU scores for two levels:

- *Sentence-Level BLEU Score:* Assesses the accuracy of each sentence created by the model against its respective reference sentence from, *The Gift of the Magi*. This aids in the evaluation of the generated text's local coherence.
- *Corpus-level BLEU score:* This metric refers to measur- ing the BLEU score over the entire generated contents against the original literary text in its entirety. It gives a more global view of the fluency and structure of the narrative being generated.

TABLE II. SENTENCE-LEVEL BLEU SCORES FOR SELECTED SENTENCES.

Sentence	BLEU Score
Jim sold his watch to buy a comb for Della	0.65
Della cut her hair to buy Jim a chain.	0.59
It was a gift of the Magi.	0.71
Della admired the beauty of her hair as she prepared for her sacrifice.	0.60
Jim pondered deeply before deciding to sell his prized watch.	0.63
Their mutual sacrifices revealed the true spirit of love.	0.68
In the end, irony reigned over their selfless acts.	0.66

Key metrics for evaluation include:

- *BLEU-1:* Measures unigram precision (single-word matches).
- *BLEU-2:* Measures bigram precision (two-word sequence matches).
- *BLEU-3 and BLEU-4:* Evaluate longer sequences, cap- turing more contextual accuracy.

C. Experimental Results and Analysis

The BLEU scores from our experiments provide insights into how well the quantum-enhanced model generates text that aligns with the original literary style.

1) Sentence-Level BLEU Scores: BLEU-1 scores are rela-tively high, indicating that the generated words closely match the reference. However, BLEU-2 and BLEU-3 scores drop as phrase lengths increase, suggesting that while individual words are correctly generated, longer sequences do not always capture the full context.

2) Corpus-Level BLEU Score: Overall, these results sug-gest that classical transformers and finetuned GPT-3 mod- els outperform the Quantum-Enhanced Model (QGAN). The BLEU-4 score values of 0.18 show that coherent and fluent sequence generation at a greater length has challenges likely attributed to quantum noise and instability during training, whereas the overall performance of the baseline transformer leads to better performance.

In our experiments, the generated text is compared to reference segments from The Gift of the Magi using both sentence-level and corpus-level BLEU evaluations.

V. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, we present the experimental results obtained with our Quantum Generative Adversarial Network (QGAN) model on The Gift of the Magi, including evaluation via BLEU score and general remarks on model limitations. We aim to evaluate the efficiency of quantum-enhanced feature extraction in text generation and scrutinize the existing challenges and bottlenecks of quantum-assisted natural language processing (NLP).

A. A. BLEU Score Evaluation

Analysis of BLEU Score Results:

Our preliminary BLEU score evaluations indicate that while the Quantum-Enhanced Language Model (QGAN) generates syntactically coherent text, it fails to fully capture the nuanced semantic depth of the text.

Key Observations:

TABLE III. BLEU SCORE COMPARISON ACROSS MODELS.

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4
QGAN (Quantum-Enhanced)	0.58	0.41	0.29	0.18
Classical Transformer	0.73	0.61	0.50	0.38
GPT-3 Fine-Tuned	0.89	0.82	0.75	0.68

- BLEU-1 Scores Indicate Basic Word Accuracy:The QGAN model's BLEU 1 score turned out to be 0.58, so it is a decent success on a word level but far less successful than classical transformers. This could indicate that the quantum generator does not destroy the selection of individual words, but fails to put them together in a reasonable order on longer sequences.
- BLEU-2 and BLEU-3 Scores Show Decreasing Accuracy: The decrease in BLEU scores for longer n-grams (0.41 for BLEU-2, 0.18 for BLEU-4) shows that while the words themselves are often well predicted, the ordering of contextually-relevant words into phrases is not yet well achieved. When comparing with the classical transformer baseline, QGAN is outperformed across all n-gram levels, suggesting that similar distributions, enabled by quantum components, currently do not confer any advantage for fluency.
- BLEU-4 Scores Highlight a Loss of Narrative Coherence: The BLEU-4 is particularly low (0.18) which indicates the QGAN model fails to maintain long-range coherence of sentences. This results in a BLEU-4 score of 0.68 for the fine-tuned GPT-3 model, demonstrating a wide margin in comparison.
- Challenges in Capturing Symbolic and Emotional Sub-tleties: The Gift of the Magi is rich in symbolism and irony, as- pects that require deep contextual awareness. Our results indicate that the QGAN model does not fully capture the deeper literary meaning present in the text.

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VI. Conclusion

This case study explored the application of a hybrid quantum-classical language model for text generation, specifically in the context of O. Henry's The Gift of the Magi. By incorporating a quantum generator within a transformer-based architecture, the model aimed to leverage quantum-enhanced representations to capture deeper narrative structures and abstract linguistic features. Our evaluation using BLEU scores, however, revealed that despite the theoretical promise of quantum computing in NLP, current quantum-enhanced models remain inferior to their classical counterparts in prac- tical performance.

The study illustrates both the potential and the shortcomings of quantum-enhanced text generation. Although quantum cir- cuits add representational power via superposition and entan- glement, their combination with traditional neural architectures remains an issue.

Despite these limitations, this study contributes to the emerging field of quantum NLP, providing a foundation for future improvements.

VII. Future Research Directions

The quantum language models are still not competitive, though they have the potential for the long term since both quantum hard and software should be in good shape in the longer term [20]. The following directions should be considered in future research and improvements in hybrid quantum-classical text generation models:

A. Improvement of Quantum Circuit Design

- Larger number of qubits: Future quantum models need to exploit larger quantum circuits that can operate on richer representations for features [13].

B. Reducing Noise and Improving Hardware Stability

- Error mitigation strategies: Leveraging quantum error correction techniques may help reduce decoherence issues and improve model consistency.

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